Prabhavit Gupta

Jr/Mid AI Engineer | Agentic AI, AWS Bedrock Agents, Databricks, LangGraph & RAG

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Executive Summary

- **Agentic AI Development & Generative AI Engineering:** Hands-on AI Engineer with 3 years commercial software and data engineering experience. Proven expertise designing and releasing LLM-based agentic systems, autonomous multi-turn workflows, and production RAG pipelines utilizing AWS Bedrock Agents, Databricks, and modern agent orchestration frameworks (LangGraph, LangChain).
- **Rigorous Prompt Engineering & Output Validation:** Expert in drafting, iterating, and benchmarking complex prompt architectures for autonomous agents, implementing strict schema validation (Pydantic), hallucination detection, guardrails, and automated evaluation suites measuring context relevance, answer accuracy, and latency.
- **Cloud Data Pipelines & Robust Python Engineering:** Solid foundation in Python, SQL, and cloud data architecture across AWS (Lambda, S3, Glue) and Databricks (PySpark, Delta Lake). Strong track record building scalable ingestion pipelines, vector indexing (pgvector, Qdrant), and production APIs with FastAPI and Docker.

Technical Skills

- **Generative AI & Agentic Systems:** AWS Bedrock (Bedrock Agents, Knowledge Bases, Claude 3.5 Sonnet / Haiku), Databricks Genie, LangGraph, LangChain, LlamaIndex, Multi-Agent Collaboration, Autonomous Tool Calling (ReAct), Prompt Engineering & Optimization.
- **Data Engineering & Cloud Platforms:** AWS (S3, Lambda, Glue, IAM, CloudWatch, API Gateway), Databricks (PySpark, Delta Tables, MLflow), Data Ingestion Pipelines, Schema Modeling, PostgreSQL, SQL Server.
- **Retrieval-Augmented Generation (RAG):** Hybrid Retrieval (Dense Embeddings + Sparse BM25), Cross-Encoder Reranking, Vector Databases (pgvector, Qdrant, ChromaDB), Semantic Chunking, Metadata Enrichment.
- **Software Development & Backend:** Python (FastAPI, AsyncIO, PyDantic, PyTest), TypeScript, Node.js, RESTful APIs, Git, Docker, GitHub Actions CI/CD.
- **Evaluation & Monitoring:** RAGAS Framework, Prompt Regression Testing, Model Observability, Cost & Latency Profiling.

Professional Experience

AI Solutions Engineer & Agentic Architect • *Independent / Applied AI Consultancy*
2024 – Present • Venice & Milan, IT

- Architected and deployed agentic workflows combining AWS Bedrock foundation models with LangGraph orchestration, automating complex multi-step document synthesis and analytical tasks for enterprise clients.
- Configured autonomous tool execution mechanisms enabling agents to interact safely with external SQL databases and REST APIs, enforcing structured JSON schemas and error-recovery handlers.
- Built resilient data ingestion pipelines on AWS (S3, Lambda, Python) and Databricks to clean, chunk, and index unstructured corporate documents into vector stores, reducing processing cycle times by 45%.
- Designed systematic prompt evaluation frameworks measuring precision, recall, and factual faithfulness (RAGAS), iterating prompts to eliminate hallucinations and achieve 94% task accuracy.
- Collaborated with client product owners, consultants, and technical stakeholders to demonstrate prototype iterations, establish requirements, and transition pilot systems into stable production.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Engineered high-throughput microservices using Python (FastAPI) and TypeScript to support automated data extraction, NLP classification, and real-time inference tasks.
- Implemented relational data models in PostgreSQL and asynchronous task processing using message queues, ensuring robust data handling for high-volume transactions.
- Developed and maintained data pipelines connecting internal databases with analytics dashboards, providing transparent operational metrics for cross-functional teams.
- Followed Agile development practices, writing clean, documented Python code with comprehensive unit and integration test coverage in PyTest.

Key Engineering Projects

- **Enterprise Agentic Assistant on AWS Bedrock:** Designed a domain-specific agent utilizing AWS Bedrock Claude 3.5 Sonnet, AWS Lambda action groups, and an OpenSearch vector knowledge base to handle multi-turn inquiries with grounded citations.
- **Automated Databricks & PySpark Ingestion Pipeline:** Built an automated batch ingestion framework in

PySpark on Databricks to process and structure multi-format data, creating normalized tables for downstream analytics and generative AI indexing.

Education & Languages

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Core coursework in formal logic, decision systems, game theory, and quantitative data analysis.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working professional proficiency).
- **Legal & Work Authorization:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted work authorization across Italy and the EU without visa sponsorship; Partita IVA registered.